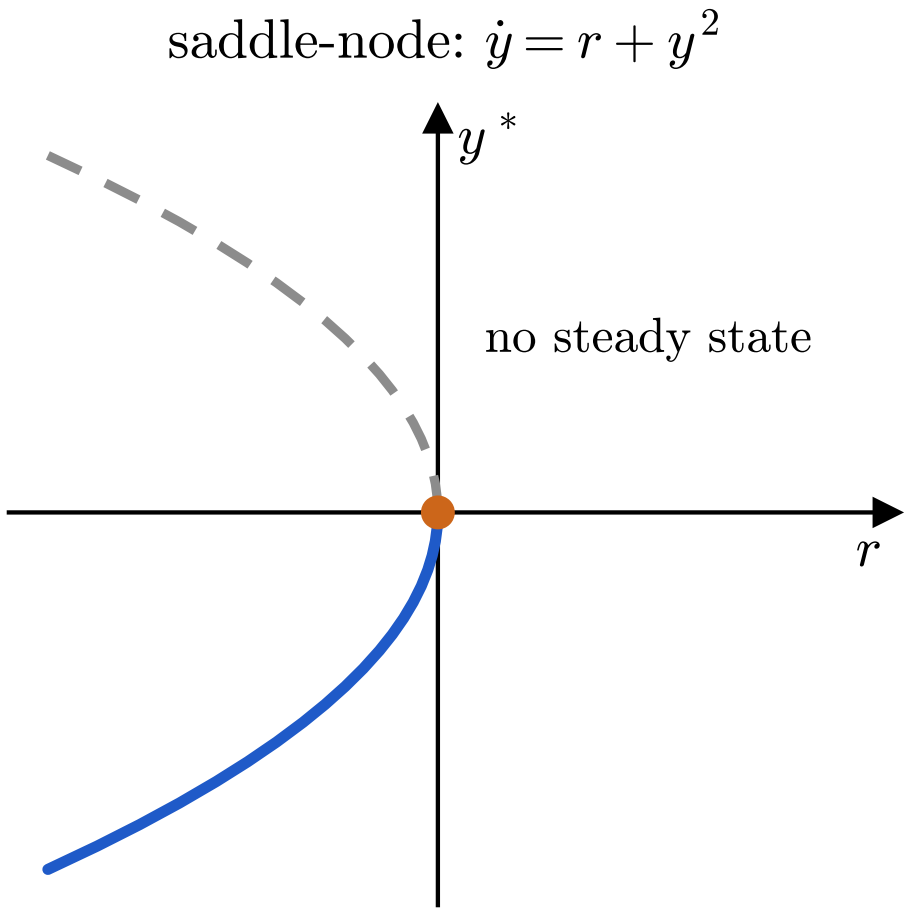
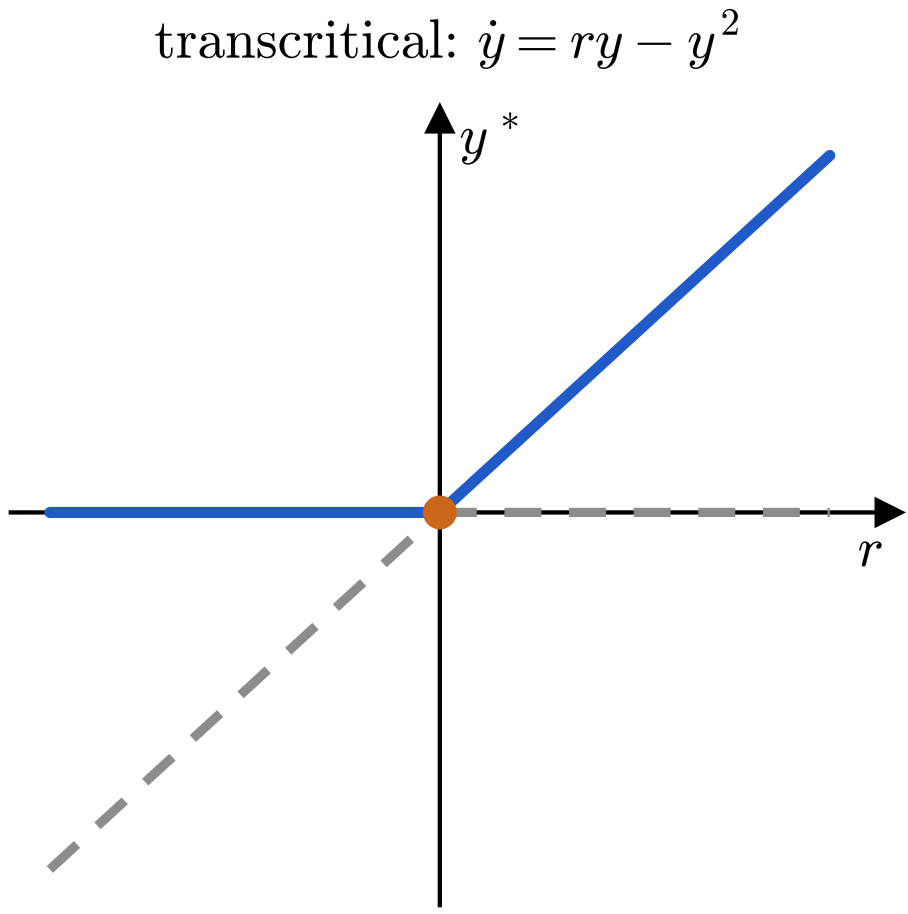

saddle-node: $\dot{y} = r + y^2$
y^*
no steady state
r


transcritical: $\dot{y} = ry - y^2$
y^*
r